

Symbolic python as a calculator

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Use python. Wip.

We assume the presence of a Jupyter installation with SymPy.

Addition

+ calculates the sum of two numbers.

$$1 + 1 = 2$$

```
x = 1 + 1  
display(x)
```

Subtraction

- calculates the difference between two numbers.

$$103 - 101 = 2$$

```
x = 103 - 101  
display(x)
```

Multiplication

* calculates the product of two numbers.

$$7 \times 9 = 63$$

```
x = 7 * 9  
display(x)
```

Division

/ calculates the difference between two numbers.

$$\frac{10}{2} = 5$$

```
x = 10 / 2  
display(x)
```

Square root

`sqrt()` calculates the square root of a number.

$$\sqrt{2} = 1.4142135623731$$

```
x = sp.sqrt(2)
display(sp.N(x))
```

Absolute value

`Abs()` returns the absolute value of a number.

$$|-7| = 7$$

```
x = sp.Abs(-7)
display(x)
```

Pi

`pi` represents the mathematical constant π .

$$\pi = 3.14159265358979$$

```
x = sp.pi
display(sp.N(x))
```

Euler's number

`E` represents the mathematical constant e .

$$e^2 = 7.38905609893065$$

```
x = sp.E2
display(sp.N(x))
```

Sine

`sin()` calculates the sine.

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

```
x = sp.sin(sp.pi / 6)
display(x)
```

Cosine

`cos()` calculates the cosine.

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

```
x = sp.cos(sp.pi / 3)
display(x)
```

Tangent

`tan()` calculates the tangent.

$$\tan\left(\frac{\pi}{4}\right) = 1$$

```
x = sp.tan(sp.pi / 4)
display(x)
```

Arcsine

`asin()` calculates the inverse sine.

$$\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

```
x = sp.asin(sp.Rational(1, 2))
display(x)
```

Arccosine

`acos()` calculates the inverse cosine.

$$\arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

```
x = sp.acos(sp.Rational(1, 2))
display(x)
```

Arctangent

`atan()` calculates the inverse tangent.

$$\arctan(1) = \frac{\pi}{4}$$

```
x = sp.atan(1)
display(x)
```

Exponential

`exp()` calculates the exponential function e^x of a number.

$$e^3 = 20.0855369231877$$

```
x = sp.exp(3)
display(sp.N(x))
```

Logarithm

`log()` calculates the logarithm.

$$\ln(10) = 2.30258509299405$$

```
x = sp.log(10)
display(sp.N(x))
```

Exact fractions

`Rational()` represents a fraction in exact form.

$$\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

```
x = sp.Rational(1, 3) + sp.Rational(1, 6)
display(x)
```

Simplification

`simplify()` attempts to transform an expression into a simpler form.

$$\sin^2(x) + \cos^2(x) = 1$$

```
x = sp.symbols('x')
expr = sp.sin(x)**2 + sp.cos(x)**2
result = sp.simplify(expr)
display(result)
```

Expansion

`expand()` expands an expression.

$$(x + 1)^3 = x^3 + 3x^2 + 3x + 1$$

```
x = sp.symbols('x')
expr = (x + 1)3
result = sp.expand(expr)
display(result)
```

Factorization

`factor()` factors a polynomial into the product of its factors.

$$x^2 - 1 = (x - 1)(x + 1)$$

```
x = sp.symbols('x')
expr = x2 - 1
result = sp.factor(expr)
display(result)
```

Equations

`solve()` solves an equation or a system of equations.

$$x^2 - 5x + 6 = 0 \quad x = 2 \quad x = 3$$

```
x = sp.symbols('x')
solutions = sp.solve(x2 - 5*x + 6, x)
for solution in solutions:
    display(solution)
```

Derivatives

`diff()` calculates the derivative of an expression.

$$\frac{d}{dx}(x^3) = 3x^2$$

```
x = sp.symbols('x')
result = sp.diff(x3, x)
display(result)
```

Integrals

`integrate()` calculates the integral of an expression.

$$\int x^2 dx = \frac{x^3}{3}$$

```
x = sp.symbols('x')
result = sp.integrate(x**2, x)
display(result)
```

Limits

`limit()` calculates the limit of an expression.

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

```
x = sp.symbols('x')
result = sp.limit(sp.sin(x) / x, x, 0)
display(result)
```

Summations

`summation()` calculates the sum of an expression.

$$\sum_{k=1}^{10} k = 55$$

```
k = sp.symbols('k')
result = sp.summation(k, (k, 1, 10))
display(result)
```

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